

The Particle Model — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says explain, your answer must talk about particles — their arrangement, their movement, the spaces between them, or the forces between them.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe experiments you may not have seen before. You are not expected to recognise them — apply the particle ideas you already have.
- Never write that particles melt, expand, or get bigger. Only their arrangement, spacing and energy change.

SECTION 1 – EVAPORATION AGAINST BOILING

- 1.** Give four differences between evaporation and boiling. Each difference must say something about both processes. [4]

- 2.** Water boils at 100 °C. Explain how a puddle can still disappear completely on a day when the temperature never rises above 9 °C. [3]

- 3.** A wet cloth wrapped around a bottle keeps the bottle cool. Explain why evaporation [3]
cools the liquid that is left behind.

- 4.** State three things that would make water evaporate faster, and explain one of them [4]
in terms of particles.

SECTION 2 – GAS PRESSURE

- 5.** A sealed metal tin containing air is placed in a hot oven. Explain, in terms of [3]
particles, why the pressure inside the tin increases.

- 6.** A gas in a sealed syringe is pushed into half its original volume, at the same [3]
temperature. Explain what happens to the pressure, and why.

SECTION 3 – UNFAMILIAR EXPERIMENTS

- 7.** A drop of liquid bromine is released at the bottom of a tall sealed jar. In one
experiment the jar contains air; in another the air has been pumped out first. The
brown bromine vapour fills the empty jar almost instantly, but takes several
minutes to fill the jar containing air.

(a) Name the process by which the bromine spreads. [1]

(b) Explain why it is so much slower in the jar containing air. [3]

8. Washing dries faster on a windy day than on a still day of the same temperature. [3]
Explain why.

9. A burn from steam at 100 °C is far more serious than a burn from boiling water at 100 °C, even though both are at the same temperature. Explain why. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.